

BARRAGE ANGLIERS, QC, Canada

WMO: 718201

Lat: 47.550N

Lon: 79.240W

Elev: 267

StdP: 98.16

Time Zone: -5.00 (NAE)

Period: 96-07

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.0	-27.0	-34.0	0.2	-29.4	-30.9	0.2	-26.7	8.4	-12.1	7.7	-11.5	1.6	90	0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	29.6	20.3	27.9	19.2	26.1	18.3	21.8	27.5	20.7	25.7	19.7	24.0	3.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	15.2	24.1	19.0	14.2	23.2	18.1	13.4	22.5	65.2	27.3	61.2	25.9	57.7	24.0	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.3	6.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.1	-13.7	-11.3	-5.9	3.0	11.1	17.1	18.8	17.7	14.0	6.4	-0.8	-7.6		
	DBStd	12.56	7.64	6.92	7.03	5.13	4.96	4.55	3.37	3.45	4.15	4.78	4.81	6.66		
	HDD10.0	3107	736	597	492	215	49	4	0	0	10	136	324	546		
	HDD18.3	5319	994	831	750	460	229	76	35	54	141	372	573	804		
	CDD10.0	969	0	0	0	5	82	218	272	237	131	23	1	0		
	CDD18.3	139	0	0	0	0	4	39	49	34	12	1	0	0		
	CDH23.3	1149	0	0	0	1	61	349	410	244	83	2	0	0		
	CDH26.7	291	0	0	0	0	14	94	108	58	16	0	0	0		
(14) Wind	WSAvg	3.0	2.9	3.0	3.2	3.2	3.2	2.8	2.7	2.7	2.9	3.3	3.3	3.0		
(15) Precipitation	PrecAvg	964	63	51	60	64	80	100	102	107	102	84	80	73		
	PrecMax	1184	116	94	126	110	152	211	197	234	208	156	159	117		
	PrecMin	752	15	15	11	16	35	38	29	37	41	27	10	31		
	PrecStd	115	21	24	27	27	34	40	40	48	38	32	34	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.5	6.6	12.8	21.5	29.0	31.1	32.1	30.9	28.7	23.2	12.4	6.4		
		MCWB	4.6	2.4	5.3	12.6	18.3	21.2	22.3	21.2	19.8	16.9	10.5	3.9		
	2%	DB	2.2	3.1	9.3	17.6	24.7	29.4	29.3	28.3	25.6	19.3	10.8	4.1		
		MCWB	1.2	0.5	5.4	9.0	14.0	20.0	20.1	20.1	18.9	15.1	9.4	2.8		
	5%	DB	0.3	1.4	6.9	14.2	22.5	27.4	27.5	26.1	23.2	15.9	8.4	2.4		
		MCWB	-0.3	-0.1	4.1	7.5	13.6	18.5	19.1	18.8	17.6	12.9	7.2	1.4		
	10%	DB	-1.9	-0.2	4.5	11.1	19.8	25.2	25.6	24.1	21.0	13.4	5.7	1.0		
		MCWB	-2.6	-1.1	1.7	6.2	12.9	17.8	18.2	17.8	16.5	11.1	4.3	0.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.1	2.3	8.9	14.1	19.1	22.3	23.4	23.1	20.9	18.2	11.6	5.4		
		MCDB	5.5	5.5	9.6	20.0	24.6	29.3	28.3	28.0	26.2	21.4	12.2	5.9		
	2%	WB	1.4	0.9	6.1	10.4	16.8	21.2	22.1	21.4	19.7	15.6	9.2	2.8		
		MCDB	2.0	2.6	8.1	14.8	22.3	27.1	27.6	26.1	24.1	18.9	10.4	3.8		
	5%	WB	-0.3	0.0	4.1	8.2	15.0	20.1	20.7	20.2	18.6	13.3	7.1	1.3		
		MCDB	0.4	1.2	6.9	13.8	20.1	25.2	25.6	24.0	22.4	15.7	8.4	2.3		
	10%	WB	-2.6	-1.3	2.0	6.1	13.3	18.9	19.4	19.0	17.0	11.2	4.4	0.3		
		MCDB	-1.9	-0.1	4.1	10.3	18.3	23.5	23.4	23.0	20.2	13.2	5.6	1.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	11.8	12.1	10.5	10.8	10.4	9.9	9.4	9.2	7.1	5.9	8.2		
		MCDBR	9.9	11.1	12.4	16.3	15.0	13.7	12.7	12.3	12.2	11.4	8.9	6.9		
	5% WB	MCWBR	9.5	9.2	8.8	9.3	6.8	6.1	5.3	5.4	6.4	7.8	8.0	6.5		
		MCWBR	9.7	10.0	10.7	15.0	12.4	11.5	11.1	10.0	10.8	10.9	8.7	6.9		
(40) Clear-Sky Solar Irradiance	taub		0.265	0.284	0.315	0.355	0.361	0.389	0.395	0.384	0.364	0.329	0.303	0.264		
	taud		2.276	2.215	2.237	2.323	2.395	2.343	2.354	2.378	2.413	2.497	2.450	2.350		
	Ebn at Noon		817	876	902	898	902	875	863	858	843	824	773	777		
	Edh at Noon		75	100	115	118	114	121	118	111	97	77	65	62		
(44) All-Sky Solar Radiation	RadAvg		1.22	2.26	3.77	4.67	5.22	5.64	5.53	4.65	3.52	2.09	1.09	0.87		
	RadStd		0.18	0.34	0.41	0.40	0.45	0.38	0.46	0.32	0.35	0.28	0.12	0.09		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page